Des-Lux Play

Part 3: Fixed Mistakes, Updates, Modifications and JavaScript Additions

Enhancing Functionality and SEO

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1. Overview

This report lists, in one place, everything that was fixed, updated, modified and added between Part 2 and the finished Part 3 website. It is organised as: fixed mistakes (section 2), updates and modifications by file (sections 3 and 4), everything added in JavaScript (section 5), new images (section 6), testing (section 7) and what is still to do (section 8).

Measure	Result
Mistakes fixed	6 open items from Part 2, 6 further corrections found in Part 2 work, and 3 bugs found and fixed while building Part 3
Files modified	5 HTML pages, css/style.css, README.md
Files added	404.html, robots.txt, sitemap.xml, 6 JavaScript files, 6 README screenshots, 3 product photographs
JavaScript	6 files, 1,286 lines, about 50 KB (unminified, with comments)
Images	12 images, about 1.1 MB in total (the original nine PNGs were 16.2 MB)
Gallery	7 photos in Part 2 order, now 10 with the three new product photographs
Tests	53 automated browser checks passed; no broken links; no duplicate IDs; no sideways scrolling at 390 px

Part 2 home page, then the Part 3 products page:

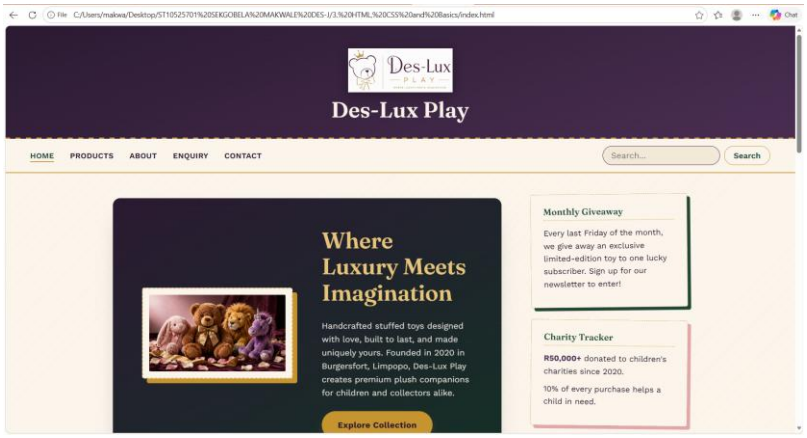


Figure 1. Part 2 home page

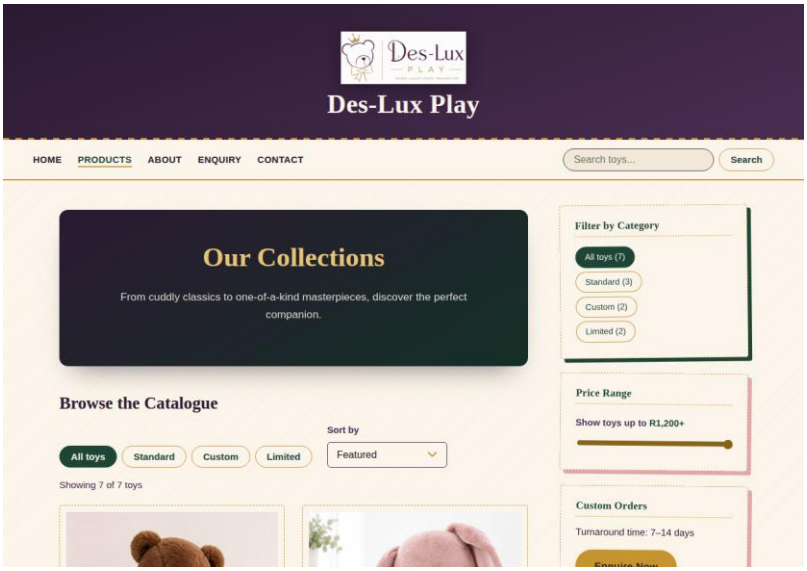


Figure 2. Part 3 products page with seven toy photographs, filters, sort and price slider

Screenshots were taken in a test browser without internet access, so Google Fonts show as the fallback serif.

2. Fixed Mistakes

2.1 Open items listed in the Part 2 README

Mistake	Fix	Files
Search box was markup only and did nothing	Now a real GET form. On products.html it filters live as you type; on other pages it opens products.html?q=... Empty searches are blocked.	all pages, site.js, products.js
Newsletter signup was markup only	Now validated, uses a honeypot, sends by AJAX and shows a status message.	index.html, site.js
Category filter links were href="#" and did nothing	Links open products.html?category=... and filter instantly. Sidebar counts now come from the real data (Standard 3, Custom 2, Limited 2).	products.html, products.js
Forms posted to a placeholder with no checking	Forms validate in the browser and submit by AJAX. The placeholder remains until a Formspree ID is added; the site then runs in a visible demo mode.	enquiry.html, contact.html, forms.js, config.js
Map embeds were approximate Google iframes	Replaced by a Leaflet map built from coordinates in the page, plus an Open in Google Maps link on each location.	contact.html, map.js
Image filenames contained typos (standdard, uicoron, fouder)	All images renamed with descriptive names and every reference updated.	images/, all pages

2.2 Further corrections found in the Part 2 work

Mistake	Fix	Files
The brand name was the h1 on every page and h2 was skipped	One h1 per page, sections are h2, small headings h3. The header brand is now a styled div.	all pages, style.css
Logo width and height (500 x 150) did not match the real image	Corrected to 500 x 209 so the page does not jump while loading.	all pages
Gold text was hard to read on ivory (poor contrast)	Added a darker gold (--gold-dark) and used it for gold text.	style.css
Images were 16.2 MB in total	Converted to WebP and resized; now about 1.1 MB for twelve images.	images/
Fonts loaded through a CSS @import, which delays text	Moved to <link> tags with preconnect and display=swap.	all pages, style.css
README screenshot path (screenshots/) did not match the folder tree (docs/screenshots/)	Paths now match and the screenshots were retaken.	README.md, docs/

2.3 Bugs found and fixed while building Part 3

Bug	Fix	File
The Leaflet script tag on the contact page was treated as a file name, so the map library never loaded (found by browser testing)	Script tags are now written out correctly, and the map was tested again.	contact.html
The newsletter form showed the browser's own error bubble instead of the site's message	Native validation is switched off by script (noValidate), while the HTML required attribute still protects visitors without JavaScript.	site.js
The enquiry meta description was 163 characters, over the 160 limit	Shortened to 149 characters.	enquiry.html

3. Updates

- **Catalogue:** the three static product sections became a generated catalogue of seven toys. Custom Creations is now two entries (dragon and unicorn).
- **Photographs:** the Velvet Bunny, Cuddle Elephant and Silver Fox now have photographs, so no toy shows a placeholder tile.
- **Gallery:** the About page gallery grew from 7 to 10 photos.
- **Enquiry form:** new Needed-by date, field hints and a result panel with an estimate, availability and deposit.
- **Contact form:** optional phone number, minimum message length, character counter and a compiled-email result panel.
- **Contact page:** a Find my nearest store button, two location cards and an interactive map.
- **Footer:** repeats the five links plus Instagram on every page.
- **Navigation:** the current page is marked with aria-current, and a skip link was added.
- **README:** rewritten with features, SEO notes, security, deployment steps, a before-going-live list, the changelog and references.
- **Repository:** HTML files moved to the repository root so GitHub Pages can serve index.html.

3.1 SEO updates

- Unique title (29 to 54 characters) and meta description (110 to 160) on every page, with keywords, author, robots and canonical tags.
- Open Graph and Twitter card tags so shared links show a preview.
- JSON-LD structured data: Organization and WebSite (with site search) on home, Organization on About, an ItemList of Products on Products, FAQPage on Enquiry, and two Stores on Contact.
- Added robots.txt, sitemap.xml and 404.html.
- Descriptive image names and alt text, in-text internal links and a footer menu.
- Keyword research table and an off-page SEO plan (backlinks, social media, local SEO) in the README.

3.2 Security and speed updates

- Content Security Policy and referrer policy on every page; the Leaflet files carry Subresource Integrity hashes.

- A honeypot field on every form; length limits on every field; visitor text is inserted with textContent and never as HTML.
- External links use rel=noopener.
- WebP images, deferred scripts, lazy loading below the fold, a preloaded hero image and width and height on every image.

4. Modifications by File

File	What was modified
index.html	Single h1 in the hero; collection headings now h2; new call-to-action links (filtered products, enquiry, About); newsletter is a validated form with honeypot; charity total animates; new head with SEO, Open Graph, security and JSON-LD; hero image preloaded; WebP images.
products.html	Static sections replaced by a JavaScript-built catalogue; toolbar with category chips and sort; price slider and recently viewed card in the sidebar; quick-view modal; Good to Know tabs; noscript list; Product data for search engines.
about.html	Heading hierarchy fixed; in-text links to products and enquiry; new Gallery of 10 photos with a lightbox dialog.
enquiry.html	Form rebuilt with validation attributes, date field, hints and honeypot; result panel; FAQ accordion with 5 questions; FAQPage data; ?toy= preselects the toy type.
contact.html	Form gains optional phone, minimum length and counter; compiled-email result panel; iframes replaced by a Leaflet map, nearest-store button and two location cards; Store data for search engines.
css/style.css	Heading scale shifted for the h1/h2/h3 hierarchy; @import removed; --gold-dark added; header brand and nav search restyled; new section 14 (wide sections, entrance animation, catalogue cards, dialogs, tabs, accordion, gallery, form errors and results, map and pins, footer links, responsive and reduced-motion rules). Grew from about 16 KB to about 32 KB.
README.md	Rewritten for Part 3 (see section 3).

5. What Was Added in JavaScript

Six files were added in the new js/ folder. The pages load config.js first, then site.js, then only the files that page needs. All scripts use strict mode and plain JavaScript; the only library is Leaflet (for the map).

File	Lines	Purpose and main features
js/config.js	32	One place for the form URL, the e-mail recipient, turnaround days, estimate prices and deposit percentage. Works out demo mode automatically while the Formspree placeholder remains.
js/site.js	289	Shared code for every page: DOM builder, e-mail check, AJAX sender, header search guard, newsletter, count-up animation, accordion, tabs, gallery lightbox.
js/products-data.js	213	The catalogue as an array of seven toys (id, name, category, price, label, status, summary, description, details, image, tags). Add a toy by copying one entry.
js/products.js	265	Builds the product cards and handles search, category filter, price filter, sort, shareable URLs, the quick-view modal and recently viewed.
js/forms.js	358	Validation rules, error messages, character counters, AJAX submission, the enquiry estimate and the compiled e-mail.

File	Lines	Purpose and main features
js/map.js	129	Leaflet map with OpenStreetMap tiles, pins, popups, Show on map buttons, nearest-store finder and a fallback if the map cannot load.

5.1 js/site.js in detail

Feature	How it works
h() element builder	Creates elements with attributes and children. Text is added with createTextNode, so visitor input can never become HTML.
isEmail() and sendForm()	Shared e-mail check. sendForm() posts the form with fetch and returns a result object; it never throws. In demo mode it waits 0.6 seconds and reports success without sending.
Header search guard	On pages other than products, stops an empty search from submitting.
Newsletter	Validates the address, checks the honeypot, shows Subscribing, then a success or error message. Demo mode says so.
Count-up	An IntersectionObserver starts a requestAnimationFrame animation (ease-out, 1.4 s) when the charity total scrolls into view. The real figure is already in the HTML, and reduced-motion users see it straight away.
Accordion	Buttons with aria-expanded and aria-controls. Collapsed panels get the inert attribute. Up, Down, Home and End move between questions. Height animates with a CSS grid transition.
Tabs	Buttons with role tab. Left, Right, Home and End keys move between tabs; only the selected panel is visible.
Lightbox	Gallery buttons open a native dialog. Previous and next buttons and the arrow keys move through the 10 photos and wrap around; a counter reads for example 3 of 10. Esc, the close button or clicking the backdrop closes it.

5.2 js/products.js and js/products-data.js in detail

Feature	How it works
Card rendering	card() builds each article from the data; results are inserted through a DocumentFragment and replaceChildren.
Search	The header search box filters live. Every word typed must appear in the toy's name, summary, status, category or tags.
Category filter	Toolbar chips and sidebar links share one delegated click listener. The active one shows aria-pressed or aria-current.
Price filter	The sidebar slider (R300 to R1,200+) hides toys above the chosen price. Toys with no price yet are never hidden.
Sort	Featured, price low to high, price high to low, name. Unpriced toys always sort last.
Result line and Clear filters	Reads for example Showing 2 of 7 toys for lion, and a Clear filters button appears whenever any filter is active.
Shareable URLs	history.replaceState keeps the address as products.html?category=limited&q=lion. The page reads the same parameters when it opens.
Quick-view modal	A native dialog filled from the data, with a call-to-action that links to the enquiry form with that toy preselected.
Recently viewed	The last three toys opened are kept in localStorage (wrapped in try/catch) and listed in the sidebar.

5.3 js/forms.js in detail

Feature	How it works
Validation rules	Names: letters, spaces, hyphens and apostrophes. E-mail format. South African phone numbers (082 123 4567 or +27 82 123 4567). Needed-by date at least 7 business days ahead. Reference image must be an image of 2 MB or less. Required, minimum and maximum lengths are read from the HTML attributes.
Error display	An error line is created under each field and linked with aria-describedby; the field is marked aria-invalid and styled red. Fields validate when you leave them and re-check as you type.
Submit handling	Validates every field, focuses the first error, checks the honeypot, shows Sending, then calls sendForm() and handles success or failure.
Character counters	Live counts on the message boxes (500 for enquiry, 1,000 for contact).
Enquiry estimate	Looks up an indicative price by size, works out the 50% deposit, compares the estimate with the chosen budget, and flags a needed-by date that is inside the longest turnaround.
Result panels	The form is replaced by a panel (focus moves to it) with a button that builds a mailto: link to the recipient in config.js, and a button to start again. All text is inserted with textContent.
Prefill	?toy=dragon (and similar) preselects the toy type on the enquiry form.

5.4 js/map.js in detail

Feature	How it works
Markers from the page	Reads data-lat and data-lng from the two location cards, so the cards and the map cannot disagree.
Map and pins	OpenStreetMap tiles with attribution; custom gold stitched-pin markers drawn in CSS; popups built from DOM nodes with a Google Maps link.
Show on map buttons	Fly to the chosen store (instant move for reduced-motion users) and open its popup.
Find my nearest store	Uses the browser's geolocation and the haversine formula to find and show the closest store.
Fallback	If Leaflet cannot load, the map box explains this and the Google Maps links on each location still work.

5.5 Animations and transitions

- Banner entrance: the photograph on each page settles into its frame when the page loads (CSS keyframes).
- Count-up of the charity total (JavaScript).
- Accordion height, tab panel and dialog opening transitions; product card and gallery hover lifts.
- Every animation is switched off for visitors who prefer reduced motion.

5.6 Advanced DOM manipulation

- Elements created with createElement and inserted through DocumentFragment and replaceChildren.

- One delegated click listener serves every product card and filter control.
- IntersectionObserver for the counter; history.replaceState for shareable URLs; localStorage for recently viewed.
- Native dialog methods showModal() and close(); the inert attribute for collapsed accordion panels.

6. New Images

Source file	New file	Used in
Velvet Bunny.png	velvet-bunny-pastel-pink-plush.webp	Products catalogue and modal, About gallery
Cuddly Elephant.png	cuddle-elephant-grey-plush.webp	Products catalogue and modal, About gallery
Silver Fox.png	silver-fox-limited-edition-plush.webp	Products catalogue and modal, About gallery

Each was cropped to a square to match the other cards, converted to WebP (80 to 95 KB, down from about 2.3 MB each) and given descriptive alt text. The bunny and elephant crops cut off a little of the feet to keep the ears in frame.

7. Testing

- 53 automated browser checks passed, covering: catalogue rendering, category chips, sidebar links, search (including no results), sorting, price slider, clear filters, shareable URLs, the modal and Esc, recently viewed, tabs and arrow keys, header search from another page, newsletter errors and success, the counter, the lightbox (open, next, wrap around, Esc), accordion, enquiry and contact validation messages, past dates, wrong file types, the estimate, the mailto link, protection against injected HTML, and the honeypot.
- The map logic was tested with a stand-in for Leaflet because the test browser had no internet. The real map should be checked after deployment.
- A link checker found no broken internal links, missing anchors or duplicate IDs; every page has exactly one h1.
- Home, products and contact pages were checked at 390 px wide with no sideways scrolling.

8. Still To Do Before Submission

Item	Why
Add the Part 2 lecturer feedback to the README changelog	It was not available when this was written
Create a Formspree form and replace YOUR_FORM_ID (js/config.js and the three form actions)	Until then the forms run in demo mode and send nothing
Confirm the estimate prices in js/config.js	Only the R500 starting price came from the brief
Replace placeholder phone numbers, e-mails, addresses and map coordinates	They are stand-ins; the coordinates are approximate town centres
Keep robots.txt and sitemap.xml at the same level as index.html	Search engines only look for them at the top level (your latest zip had them in a sitemap/ folder)

Item	Why
Check the live map after deploying	Leaflet loads from the internet and could not be tested live
Add access dates to the README references and confirm the source of the three new images	
Push to GitHub, enable Pages, and submit the repository link	Submission requirement